SpeedUp v2 User Check 7.1

Human-reviewed goals. Persistent visual decisions. Measured motion fragments.

Architecture, data contracts, calibrated direction semantics, execution control and independent evaluation.



Recorded worked examples
t0: soup + tomato sauce into basket.
t1: cream cheese + butter into basket.

English technical guide · October 30, 2026
Frozen UC7.1 Version A / A9 · gpt-6-astra / high

DOCUMENT CONTRACT

Scope, release and reading map

This guide describes the frozen implementation used in the two approved LIBERO-10 attempts.

Item	Meaning
Release identity	agent-speedup-v2-user-check-7.1; direction_semantics = external_view_v1.
Framework chapters	Model/executor boundary, plans and approval, direction semantics, actions, feedback and deployment protocol.
Evaluation chapters	Official goal, terminal physical truth, diagnostic settling, timing and recorded contact accounting.
Worked examples	One rearranged layout each for t0 and t1; official init 0 / seed 7; actual browser approval.
Evidence boundary	These two runs are not a ten-task sweep, multi-seed study, baseline ablation or measured UC7.1 speedup.
Source policy	Runtime, prompts and scoring were unchanged during both runs. Old v7 deck remains a separate historical report.

[P] UC7_1_ASTRA_HIGH_TEST_PROTOCOL.md; [L] frozen release manifest; [R0,R1] approved run artifacts.

ARCHITECTURE

Three responsibilities, one control loop

The model owns task decisions; the executor owns repeated motor control; evaluation remains separate.

Task executive

Interpret images and task.
Choose target, grasp and route.
Inspect visual outcomes.
Revise method from feedback.

→

Measured executor

Track numeric pose targets.
Run explicit gripper stages.
Check arrival and progress.
Return control at a boundary.

→

Independent evaluator

Check the official goal.
Audit support and stability.
Record physical contacts.
Keep hidden truth out of policy.

The measured feedback loop is local to the robot. Visual reasoning and task-level recovery happen at model-decision boundaries.

[M1] agent policy; [M2] agent client; [M3] fragments; [M6] simulator and independent physical-truth evaluator.

LIFECYCLE

From preparation to one authorized attempt

A draft is generated before policy actions; approval is consumed once before execution requests.

Prepare

Validated layout
Frozen reset
One model draft

→

Review

Images + selectors
Check every step
Approve plan

→

Execute

Observe / decide
Measured fragments
Fresh feedback

→

Finalize

Official terminal
Physical diagnosis
Video + audit

Continuous execution: fresh RGB + measured robot state → model decision → bounded local sequence → measured result → next observation.
Authorization boundary: one approved task/layout/configuration, one attempt; no automatic episode restart.

[P §§4–6] preparation, approval and supervision; [M4] ReviewStore; [M5] user-check runner.

OBSERVATION CONTRACT

What enters the policy—and what does not

Calibration is allowed in UC7.1. It does not supply object depth or hidden task answers.

Policy-visible input

Two upright RGB views at 512×512.
Measured robot pose / gripper state and seven arm joints.
Camera and fixed tool geometry for execution.
Official task text, reviewed goals, action history and measured feedback.

Evaluator-only information

Simulator object poses and identities.
BDDL numeric goal regions and success predicates.
Physical contact, support and stability truth.
Reset validation and post-terminal diagnosis stay outside model input.

Initial plan/replan requests use images and task context without camera calibration; execution requests carry calibrated geometry.

[M2,M6] calibrated observation / executive request path; [M7] plan client; [P §2] allowed input boundary.

PLAN DATA MODEL

The reviewed plan is a semantic contract

Object references and goals are reviewed; a plan is not a fixed trajectory or a grasp certificate.

Field	Role
parts	Stable IDs + visual descriptions; x = external-grid cell, w = wrist-grid cell; move = free/fixed/hinged/sliding/unknown.
steps	do, obj, ref, grip, sense, keep, until define an ordered task intent and selected constraints.
hold_true	Relations that the plan asks to preserve across the task.
regions	Support / anchor / relation / frame / bounds_xy_m define spatial targets. Directional bounds may remain null.
placement_goals	Object/reference relation and optional tolerance; independent evaluation still owns task success.
Grid convention	8×8, 1–64 row-major from top-left; 0 means invisible. A cell identifies a region, not a metric contact point.

[M4,M7,M8] review schema, selector-edit validation and spatial extension; actual t0/t1 plan copies are preserved.

WORKED PLAN / NO MANUAL EDITS

Actual t0 plan: two objects, one container

The browser-approved draft has two place_in steps and two inside goals; extra constraints are empty.

```
parts: A=soup, B=tomato, C=basket
cells (external / wrist):
  A: 34 / 48
  B: 36 / 53
  C: 40 / 25

steps:
  place_in(A, C), sense="any"
  place_in(B, C), sense="any"

placement_goals:
  inside(A, C), tolerance=null
  inside(B, C), tolerance=null

hold_true=[], regions=[]
grip=[], keep=[], until=[] per step
```



Saved external planning grid.
No extra pose or motion trajectory was inserted by the reviewer.

[R0] planning_review/review.json and review/review.json; plan unchanged, both steps checked; shorthand rendering of exact fields.

USER CHECK INTERFACE

Browser review, edits and approval

The existing UI edits structured selectors and validates the resulting plan before approval.



Review order

Inspect both views and object cells.
Check task steps and constraints.
Save allowed selector edits.
Check every step and approve.

t0 / t1 provenance

Actual browser approval; revision 2; both steps checked; plans unchanged.
Original model drafts retained separately.

[UI,M4] actual field semantics; left panel is a labeled vector schematic, not a browser screenshot. Both runs used manual_browser_review.

INTEGRITY / SINGLE USE

Approval binds more than the JSON plan

A change to the task, initial observation, runtime, configuration or view frame invalidates the bound context.

Frozen context

Task + episode + initial RGB/state hash
Runtime SHA-256 map
Execution configuration
Preparation + view_frame

→

Approval record

Context hash + reviewed plan
Revision + checked steps
Approval timestamp
Plan hash

→

Consumption

Fresh output required
Claim marker written once
Approval cannot be replayed
Even zero-action failure consumes it

runtime_sha256 includes @configuration. A git commit alone is insufficient because the release also froze previously uncommitted files.

[M4] _context, _approval_binding, approve, claim; [M5] runtime_inventory / execution_options; [L] source manifest.

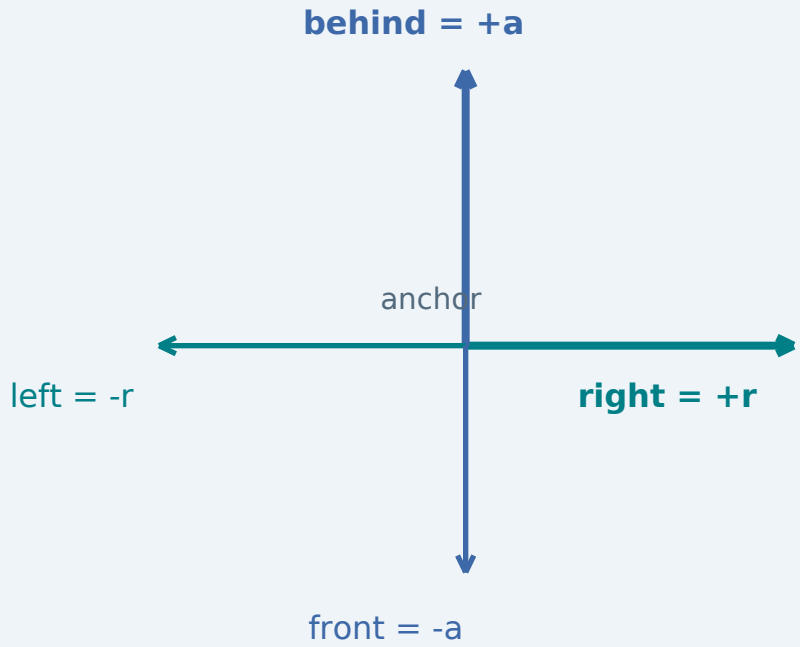
UC7.1 defines directions from the external view

The same frozen reference is used in planning language, approval, execution requests and region solving.

```
R = world-from-external-camera rotation
r = normalize(R[:2, 0])
a = (-r_y, r_x)

right_of = +r       left_of  = -r
behind   = +a       front_of = -a
```

For these recorded LIBERO scenes:
r ≈ (0, +1); **a ≈ (-1, 0)**.
World +z remains up. Numeric motion targets and region bounds still use world metres.



Global horizontal convention, not a perspective Jacobian.
Wrist images never redefine these words.

[M8] build_view_frame / region_directions; [L] A1–A3; formula uses calibration, not hard-coded LIBERO axes.

One direction source across the stack

The full task contract is sent initially; compact region_directions is included in every executive request.

Stage	Implementation contract
Preparation / approval	Compute and freeze preparation.view_frame; retain calibration, right_xy, away_xy, roll and unavailable reason.
PlanLock / policy context	Carry the frozen frame through approval consumption and the persistent episode context.
Execution request	Supply the compact direction table even when the reviewed regions list is empty.
Directional solver	For direction d, keep candidate points p with p·d greater than the anchor polygon’s maximum projection.
Unavailable / display	Missing calibration does not fall back to legacy world-axis directions in the solver. Axis-wording highlights are diagnostic only.

Large roll/inverted views remain a limitation. UC7.1 records them; it adds no image rotation, E1 preference rule or UC8 changes.

[M8] spatial solver; [M1,M2,M4,M5] propagation; [L] A0–A9 release. t0/t1 contain no directional regions, so do not validate that feature.

EXECUTIVE REQUEST / RESPONSE

Persistent model decisions between sequences

The agent keeps textual task history and a bounded visual history instead of restarting after every fragment.

Request context

Task and reviewed goal contract at initialization.
Fresh paired RGB + measured robot state.
Latest execution feedback and history.
Frozen region_directions every request.
Up to six recent visual pairs.

Decision contract

belief: explicitly model-estimated state.
plan / why: intended action and reason.
type: execute, motion, macro, observe or finish.
Only the selected payload is active.
Completed responses are validated before use.

A model “finish” claim cannot set official task success. Stable gripper gap is measured feedback, not object-attachment proof.

[M1,M2] continuous task executive and persistent history; [R0,R1] actual request/result logs with model gpt-6-astra and effort high.

EXECUTION API

Action branches and their responsibilities

Generic fragments handle repeated local control; optional native macros keep their own prerequisites.

Branch	Executed behavior	Boundary
execute	Sequence of 1–6 generic segments; at most 240 control steps per grant.	Measured completion and explicit observation checkpoints.
motion	Short numerical pose/gripper target; 1–5 local control steps.	A short grant may end before target arrival.
macro	Optional hold / turn / place_on / release path.	Its own grounding / calibration / feasibility requirements apply.
observe / finish	One pose-hold step retaining gripper command; finish also makes a model completion claim.	Neither replaces the independent benchmark predicate.

An unavailable macro returns a reason to the model; legacy grasp verification is not a universal gate before direct execution.

FRAGMENT CONTRACT

Five reusable measured-motion fragments

Coordinates refer to the robot grip site in world metres; orientation is the hand’s absolute world quaternion (wxyz).

Skill	Semantics
move_to_pose	Absolute world xyz target; quaternion_wxyz or null to retain measured entry orientation.
move_relative	Resolve xyz displacement once from the measured segment-entry pose.
approach	World displacement using the contact profile; a stall is not a contact certificate.
set_gripper	Hold entry pose; open or close as an explicit stage; require pose arrival, minimum duration and stable measured gap.
wait	Hold the entry pose and command for its declared max_steps, then return.
observation_after	Yield after that segment. Pending stages do not autoplay or resume implicitly after a handoff.

[M3] fragment_capabilities and execution engine. Skill names are generic; the model selects task-specific numerical values online.

EXECUTION EXAMPLE / T0 DECISION 3

A real grasp-and-test-lift request

The first descent stalled; the model chose to close at the reached height and inspect a short lift.

```
"execution": {"segments": [
  {"skill": "set_gripper",
   "xyz_m": null, "quaternion_wxyz": null,
   "gripper": "close", "profile": "contact",
   "max_steps": 30, "observation_after": false},
  {"skill": "move_relative",
   "xyz_m": [0, 0, 0.04],
   "quaternion_wxyz": null, "gripper": "keep",
   "profile": "contact", "max_steps": 60,
   "observation_after": true}
]}

# The 4 cm lift is model-selected,
# not a task-specific executor constant.
```



Next saved observation (step 195): soup can visibly lifted with the gripper. Attachment is judged from subsequent evidence.

[R0] recorded decision 3 at step 149; request 4 image at step 195. Payload excerpt reproduced without a new model request.

MEASURED CONTROL / 20 HZ OSC_POSE

Control bounds are local, not episode caps

Targets are tracked from measured pose with bounded Cartesian commands.

Parameter	Transit profile	Contact profile
Target translation from entry	≤ 0.30 m	≤ 0.08 m
Per-action translation cap	0.020 m	0.005 m
Position arrival tolerance	0.004 m	0.002 m
Target rotation / per-action cap	1.2 rad / 0.1 rad	0.5 rad / 0.1 rad
Rotation arrival tolerance	0.06 rad	0.04 rad

≤120 steps per segment; ≤240 per execution grant. Arrival needs two stable samples. Gripper stages require at least 8 open or 12 close actions. These local bounds do not impose a task time/action/call limit.

[M3] frozen fragment capabilities; [M5] episode limits are None. Contact profile means slower control, not guaranteed collision avoidance.

Return control when execution evidence changes

Local measurements determine completion or yield; they do not decide physical task success.

Event	Measured condition
Pose arrival	Position and rotation within profile tolerances for 2 consecutive samples; acknowledge completion only after a real action.
Gripper completion	At least 8 open / 12 close actions; 5 recent gaps span ≤0.6 mm; pose arrival also required.
Motion stall	After ≥16 actions, error improves by <0.8 mm over 12 samples. Weighted error = position error + 0.05 × rotation error.
Maintained-grip gap change	Outside set_gripper while maintaining close: gap rises ≥6 mm above its observed minimum for 3 consecutive samples.
Other handoff boundaries	observation_after, completed wait, step budget or target-envelope rejection. Pending segments never autoplay.

A stall does not certify contact identity. Pose/gap completion does not certify grasp or placement. Continuation requires a new model decision from current evidence.

[M1,M3] measured predicates and execution handoff. Thresholds are local fragment rules; they do not impose an episode-wide cap.

BENCHMARK SCOPE

LIBERO-10: the complete task registry

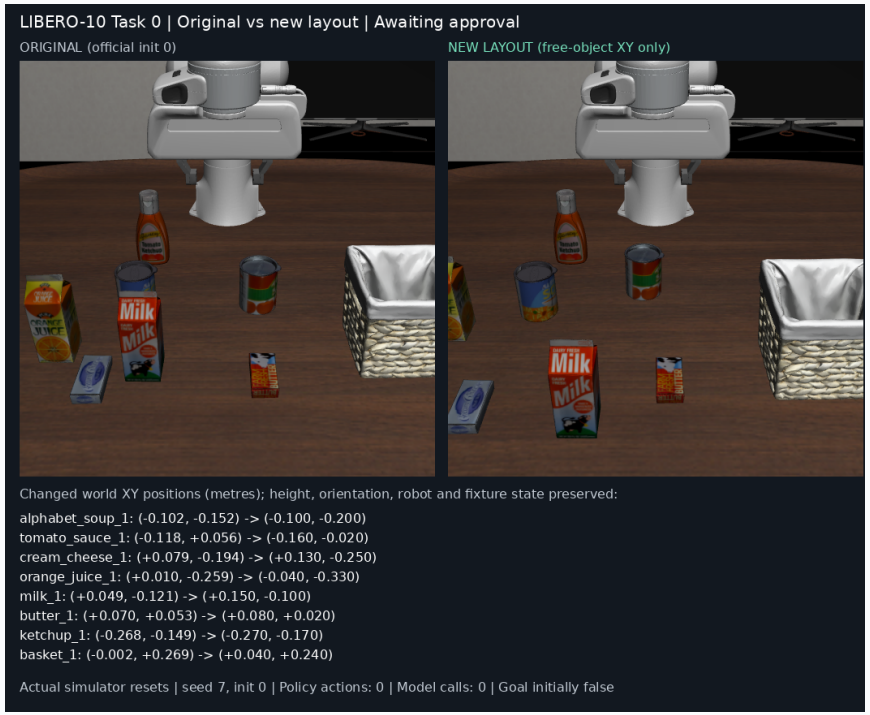
Zero-based task IDs. Only t0 and t1 are execution examples in this October 30 campaign.

ID	Official task language
t0	put both the alphabet soup and the tomato sauce in the basket
t1	put both the cream cheese box and the butter in the basket
t2	turn on the stove and put the moka pot on it
t3	put the black bowl in the bottom drawer of the cabinet and close it
t4	put the white mug on the left plate and put the yellow and white mug on the right plate
t5	pick up the book and place it in the back compartment of the caddy
t6	put the white mug on the plate and put the chocolate pudding to the right of the plate
t7	put both the alphabet soup and the cream cheese box in the basket
t8	put both moka pots on the stove
t9	put the yellow and white mug in the microwave and close it

[T] src/gpt_field/agent_speedup_v2_tasks.py:53–72; all ten exact official prompts. This guide does not claim a ten-task evaluation.

Rearranged scenes keep the task unchanged

Current protocol starts from official init 0 / seed 7 and freezes a validated custom object layout.



Actual saved t0 reset comparison: original on the left, rearranged on the right. Image content is unmodified.

Preparation checks

Move only free-object XY in the input state; retain robot, fixed objects, Z, orientation and velocity.

Preserve official task and goal logic.
Check stable support and initial goal=false.
Repeat real resets; match observation hashes.

Archive state files, change list, physical validation and both camera previews.

[P §3]; [R0,R1] layout_changes.json, preview_manifest.json, physical_validation.json and repeated-reset evidence.

OPERATING CONTRACT

Supervise one attempt without silent restarts

Each task has a separate approval, fresh output directory and one-use launch marker.

Before launch

Check frozen runtime hashes.
Verify observation + review binding.
Use an exclusive launch lock.
Keep the approved configuration.

During execution

Supervisor runs the wrapper.
No automatic process restart.
Track requests and feedback.
Contacts are record-only.

At termination

Preserve raw outcome and logs.
Run separate physical diagnosis.
Verify source and video artifacts.
Close private processes/sockets.

No task-wide time, action or call cap. Official success, existing error/conflict paths and user stop remain termination conditions. User stop is handled at the feedback boundary.

[P §§5–6] supervisor autostart=false, autorestart=false, startretries=0; [R0,R1] launch and shutdown records.

TRANSPORT / SUBSCRIPTION ONLY

Request retry is different from task retry

The 180-second deadline covers HTTP open and the full SSE response after authentication.

Condition	Frozen behavior
Overload	server_is_overloaded with HTTP 503 or absent status: at most 2 overload retries.
Transient transport	Missing completion, truncated event, stream interruption or network error: at most 2 retries in total for this category.
Retry boundary	Same observation, no response envelope and no history commit. Partial output is never executed.
Independent limits	Overload and transient counters are separate. Mixed errors can exceed 3 total sends; accepted backoff is at most 60 s.
Non-retryable / task restart	HTTP 520 / unknown_error and deadline exceeded do not retry. A new full attempt needs new approval and output.

No model downgrade or API-key fallback. t0/t1 recorded zero request errors and zero retries.

[M2] client; [X] retry classification and transport deadline; [P §6]; [R0,R1] request logs and terminal audit.

EVALUATION CONTRACT

Report four outcomes, not one SUCCESS label

The runner stops at official success; physical truth examines the state at that same boundary.

Layer	Meaning
Official LIBERO goal	task_success / libero_success use the original task predicate; an object may still be held.
Strict terminal truth	Physical checks at termination: containment, release, support and rest. Rest uses the recent 10-step displacement/rotation window when available.
Post-stop diagnosis	38 diagnostic actions: 10 open + 8 lift + 20 settle. Inspect possible release/settle effects separately.
Framework v2_success	raw_terminal AND [strict_terminal OR (hold_only AND strict_after AND raw_after)]. Read the saved frozen result; do not rescore a diagnostic pass.

Hold-only: each failing atom has only permitted hold/rest/orientation failures and also fails not_held or robot_not_touching_part. Support/below-rim failures are excluded.

[M6] independent physical-truth evaluator; [P §7]; [E] strict checks, HELD_CHECKS and terminal_failure_is_hold_only.

MEASUREMENT

Completion time and model-call accounting

Report official completion time only when official success and completion_time_eligible are true.

```
official completion time = control_elapsed_s - video_render_s
                         = control_without_video_s
```

Recorded component	t0	t1
Control-loop elapsed	532.04 s	450.99 s
Video render overhead	137.52 s	109.83 s
Official completion, video excluded	394.52 s	341.16 s
Requests: execution + reused draft	10 + 1 = 11	12 + 1 = 13

Time includes model wait, control computation, simulator actions and logging; excludes setup, planning, approval and post-stop diagnosis. It is not video playback length.

[P §8]; [R0,R1] summary.json and timing_components; retrospective removal of measured video-render overhead.

RECORDED EXECUTION / 885 POLICY ACTIONS

t0: soup and tomato sauce into the basket

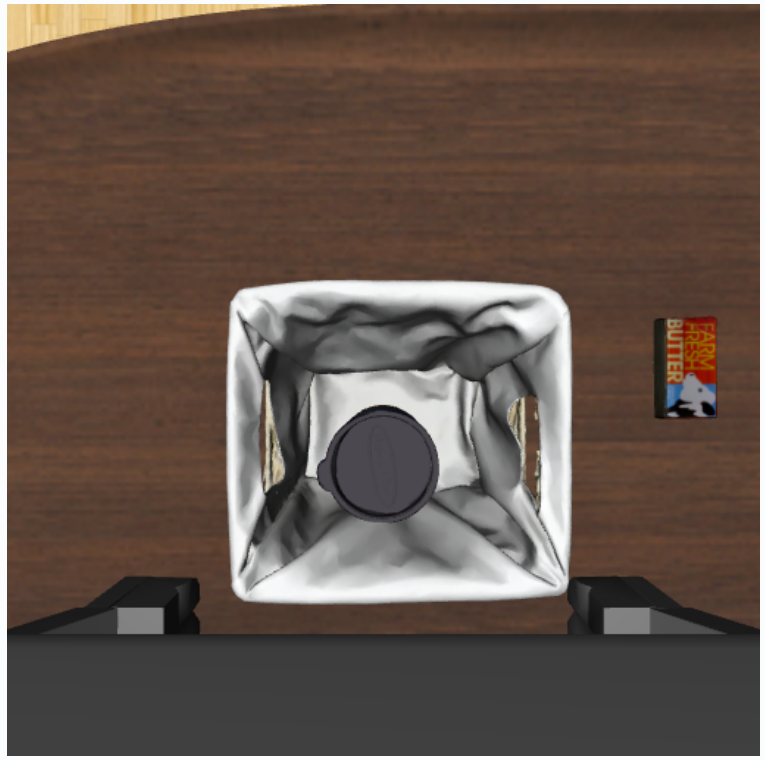
One approved rearranged attempt; 10 accepted execution decisions, all using measured fragments.



Initial external view.
Two target objects and the basket.



External view, step 195.
Soup is lifted after a short test lift.



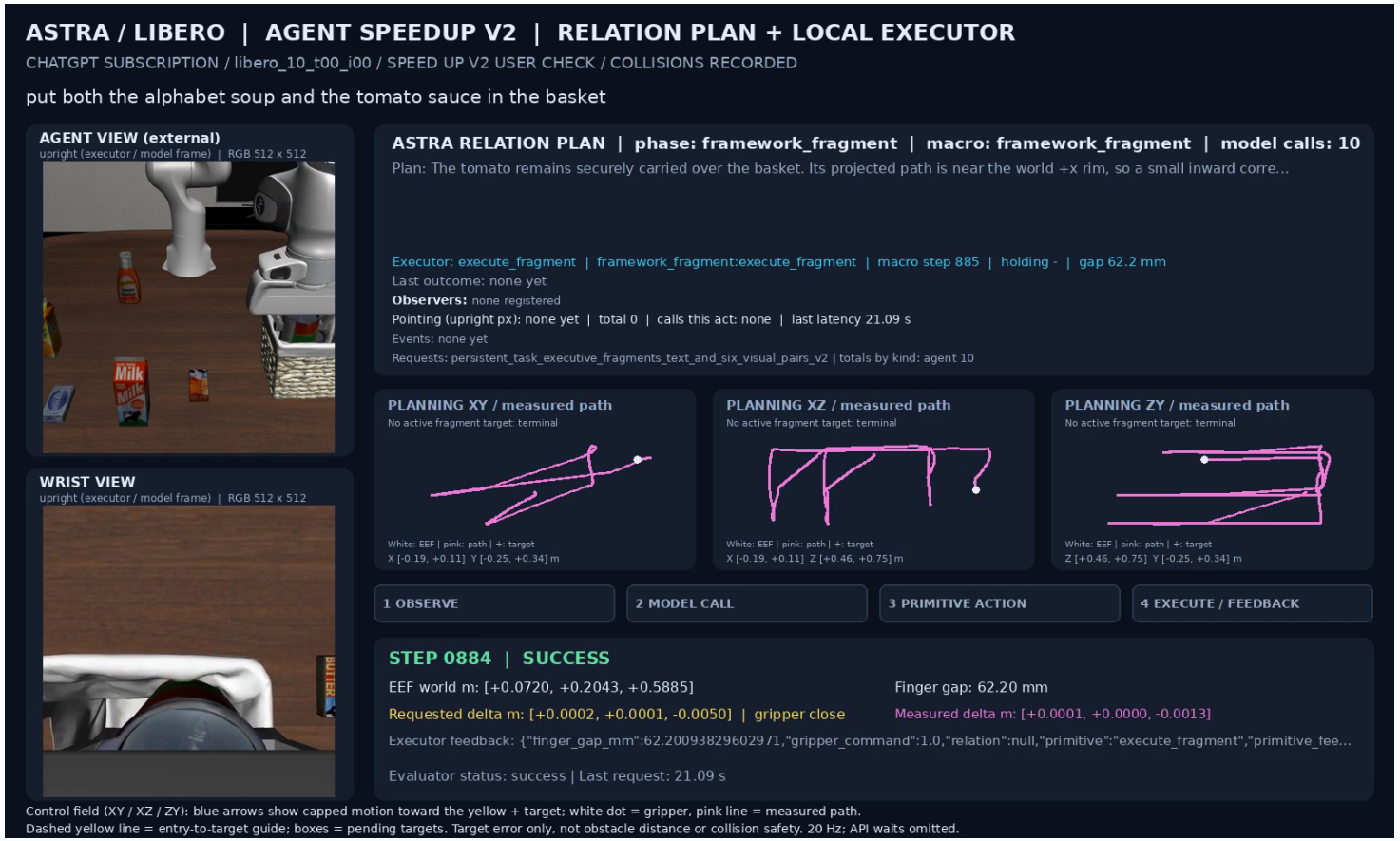
Wrist view, step 506.
Soup placement visible in the basket.

[R0] actual saved initial and request images. Policy execution proceeds next to the tomato sauce; no generated scene imagery.

TERMINAL STATE / DIAGNOSIS

t0: official success, incomplete physical placement

At termination, soup passes strict checks; tomato sauce remains held and fails support/rest checks.



Official: true
Strict terminal: false
After-settle strict: false
Framework v2: false

After diagnostic release, tomato becomes unheld and at rest, but still fails admissible container support / rim-related checks.

Its lowest point is about 15.9 mm above the container floor.

[R0] terminal_frame.png and trial/physical_truth.json. Dashboard SUCCESS denotes the official predicate, not strict physical completion.

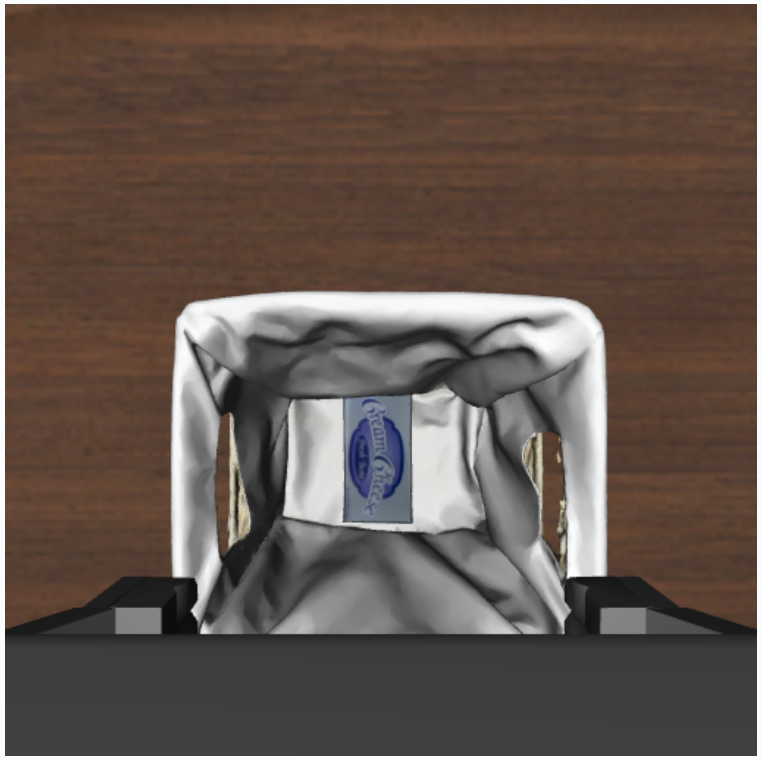
RECORDED EXECUTION / 715 POLICY ACTIONS

t1: cream cheese and butter into the basket

One approved rearranged attempt; 12 accepted execution decisions, all using measured fragments.



Initial external view.
Cheese, butter and the basket.



Wrist view, step 317.
Cream cheese placement is visible.



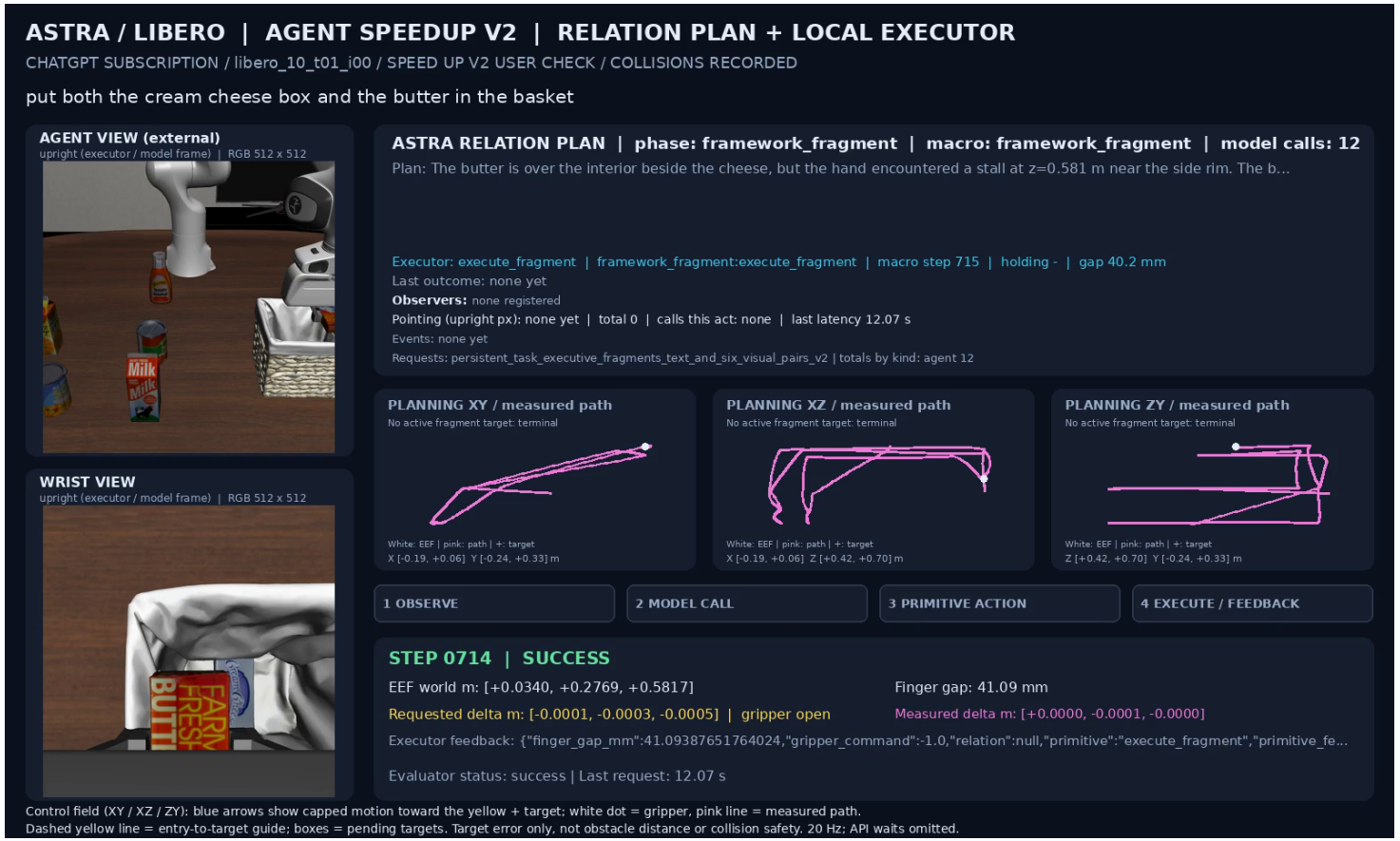
External view, step 529.
Butter is lifted for transfer.

[R1] actual saved initial and request images; terminal evaluation remains separate from these visual milestones.

TERMINAL STATE / HOLD-ONLY RULE

t1: settling helps, but does not change v2 failure

The official predicate fires during opening, after 5 opening actions and before planned withdrawal/settling.



Official: true
Strict terminal: false
After-settle strict: true
Framework v2: false

Butter fails release, rest, support and below-rim checks at termination. Cheese also fails rest.

These failures exceed the per-object hold-only exception. The later diagnostic pass cannot relabel the attempt.

[R1] terminal_frame.png, physical_truth.json and summary.v2_success; frame step 0714 is zero-based, corresponding to 715 policy actions.

Two attempts: audited outcome comparison

Both tasks reached the official predicate; neither passed the frozen framework success rule.

Measure	t0	t1
Official success	true	true
Strict at terminal	false	false
Strict after diagnostic settling	false	true
Framework v2_success	false	false
Policy actions / execution requests	885 / 10	715 / 12
Official completion, video excluded	394.52 s	341.16 s
Flagged contact samples (record-only)	11,943	4,035

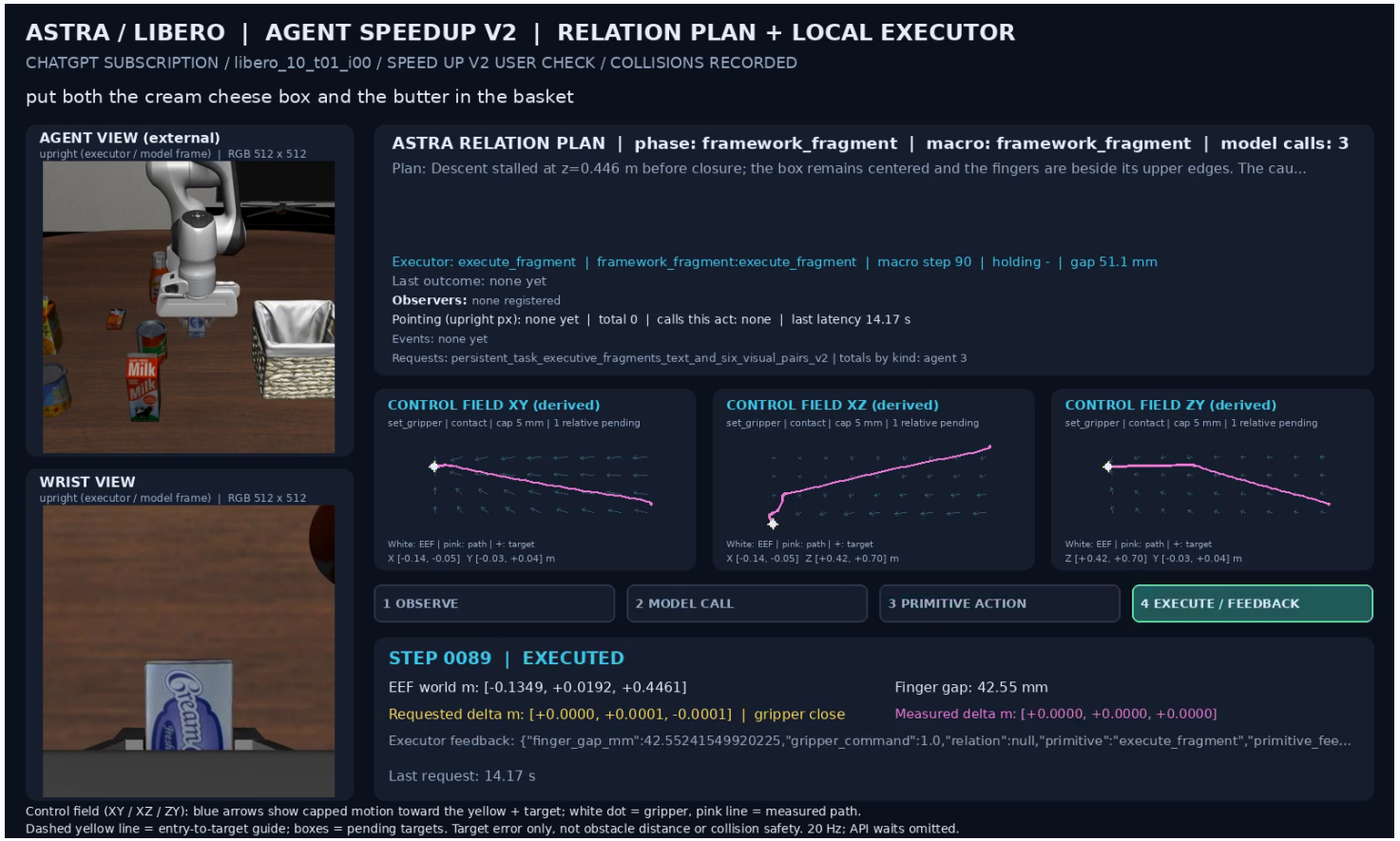
Contact samples are repeated physics observations, not counts of distinct collisions. Different tasks do not establish a speedup ratio.

[R0,R1] 23 terminal audit checks each passed; one reused planning draft each; no retry, model downgrade or API-key call.

PLANNING VIDEO / 1600×960 AT 20 FPS

Read the video as control telemetry

Saved active frame with XY, XZ and ZY projections and an embedded legend.



Blue arrows: bounded direction toward the active fragment target.
White point: gripper.
Yellow cross: active target.
Pink trail: measured motion.
Dashed yellow: entry-to-target reference.

This is a target-error control field. It does not represent obstacle clearance or certify collision safety.

[P §9]; [R1] active_frame.png. No stale target arrows at termination. API wait is compressed; playback length is not completion time.

EVIDENCE QUALITY

Validation coverage and known limits

Release tests, planning probes and the current task attempts answer different questions.

Evidence	What it establishes—and its limit
V0 release checks	350 passed, 59 pre-existing failures; 71 new checks passed, no new failures. The full regression suite was not green.
V1 planning probes	13 scenes, 13 planning calls, zero policy actions. Raw draft legality 12/13; direction/part checks 38/38.
Current campaign checks	65 focused offline checks passed; source hashes matched; 23 terminal audit checks passed for each attempt. These are integrity checks, not task success.
Current execution scope	Two single attempts on custom layouts. Both plans have regions=[]; they do not test directional-region behavior.
Unmeasured claims	No ten-task execution sweep, multi-seed robustness, matched baseline timing, collision-free guarantee or measured speedup is supplied here.

[L] release validation; [R0,R1] prelaunch / terminal audits; [E] evaluation_notes.json preserves source locations and caveats.

Reproduce the workflow with fresh approval

These are command templates, not actions performed while generating this document.

```
cd /workspace/gpt_field
RUNNER=scripts/evaluate_agent_speedup_v2_user_check.py
# Fresh TASK_DIR; verified initial_state.npy + expected observation hash.
.venv/bin/python "$RUNNER" preflight [task/layout/config flags]
.venv/bin/python "$RUNNER" prepare   [same flags] --review-dir "$TASK_DIR/review"
.venv/bin/python "$RUNNER" review --review-dir "$TASK_DIR/review" \
  --host 127.0.0.1 --port 8765
# After valid approval, invoke once through the supervised wrapper:
.venv/bin/python "$RUNNER" run --review-dir "$TASK_DIR/review" \
  --output "$TASK_DIR/trial"
```

Required preparation settings: --reasoning-effort high --planning-video --request-deadline-s 180 --stream-progress; official episode index 0, verified custom state and observation hash. Use the protocol’s environment and simulator Python. The runner config selects Astra; there is no --model flag.

The exact executable templates are in [P §5]. start --auto-approve also executes; it is not a preparation command.

Source map and reproducible document assets

Project-relative source IDs used in footers; detailed line ranges and file hashes accompany the PDF.

ID	Source or retained artifact
P / L	UC7_1_ASTRA_HIGH_TEST_PROTOCOL.md reports/uc7_direction_semantics_20261028/implementation_uc71/
M1 / M2 / M3	src/gpt_field/agent_speedup_v2_user_check_agent.py agent_speedup_v2_user_check_agent_client.py / agent_speedup_v2_user_check_fragments.py
M4 / M5	src/gpt_field/agent_speedup_v2_user_check.py scripts/evaluate_agent_speedup_v2_user_check.py
UI / E	src/gpt_field/agent_speedup_v2_user_check.html / agent_speedup_v2_user_check_server.py src/gpt_field/agent_speedup_v2_truth.py
M6 / M7	src/gpt_field/agent_speedup_v2_user_check_sim.py src/gpt_field/agent_speedup_v2_user_check_client.py
M8 / M9 / X / T	In src/gpt_field/: …_user_check_spatial.py / …_user_check_agent.py + agent_speedup_v2.py …_client.py + …_user_check_transport.py / …_tasks.py; ellipsis = agent_speedup_v2
R0 / R1	runs/agent_speedup_v2_uc71_relayout_t0_t1_20261030_001/task_0/ and task_1/ Saved trial, planning/review, approval, video, terminal audit and shutdown artifacts

Companion files: builder, evidence.json, technical_notes.json, evaluation_notes.json, exact scene images, extracted PDF text and rendering QA. Full source paths and line ranges are in the notes.